Lecture 15

Hydrogen: the spectrum

Objectives. *Lecture 14 proved that the Hamiltonian H of hydrogen is self-adjoint, that its spectrum lies in $[-\mathrm{Ry}, \infty)$ and contains $[0, \infty)$, and that every negative spectral point is an eigenvalue. We find these eigenvalues, prove their multiplicities and compare their transitions with the Balmer lines.*

- Construct the bound eigenvectors of energy $-\mathrm{Ry}/n^2$.
- Prove that their eigenspaces have dimension exactly n^2.
- Compute Balmer wavelengths and distinguish rotational from additional degeneracy.

15.1 The bound states

Lecture 14 proved that the Hamiltonian of hydrogen is self-adjoint (Theorem 14.2),

$$H = H_0 - \frac{C}{|x|} \ \text{ on } D(H) = D(H_0), \qquad a_0 = \frac{\hbar^2}{mC}, \qquad \mathrm{Ry} = \frac{\hbar^2}{2ma_0^2} \approx 13.61\,\mathrm{eV},$$

and proved what else this lecture uses. Every vector of $D(H_0)$ is a bounded continuous function (Lemma 14.1). A bounded continuous function ψ that is square integrable, twice continuously differentiable on $x \neq 0$, with bounded first partial derivatives there and a square-integrable $\Delta\psi$, lies in $D(H_0)$, and $H_0\psi = -(\hbar^2/2m)\Delta\psi$ (Lemma 14.3). Every $\psi \in D(H)$ has $\langle \psi, H\psi\rangle \geq -\mathrm{Ry}\,\|\psi\|^2$, and $\psi_1(x) = e^{-|x|/a_0}$ is an eigenvector for $-\mathrm{Ry}$ (Theorem 14.4). The essential spectrum of H is $[0, \infty)$, and every point of $\sigma(H)$ below 0 is an eigenvalue in $\sigma_{\mathrm{d}}(H)$ (Proposition 14.5 and Theorem 14.7). Every eigenvalue is negative (Proposition 14.8). What remains is to find the eigenvalues and their eigenvectors.

Constructing the bound states

Below 0 we look for eigenvectors of the form of Proposition 13.8. For $p \in \mathcal{V}_l$, the function $\psi = u(|x|)p(x)/|x|^{l+1}$ satisfies $-(\hbar^2/2m)\Delta\psi + V\psi = E\psi$ on $x \neq 0$ when $H_l u = Eu$, with $V(r) = -C/r$ in the radial rule H_l. Write $E = -\hbar^2\eta^2/2m$ with $\eta > 0$. Far out the equation is close to $u'' = \eta^2 u$, whose decaying solution is $e^{-\eta r}$, and near 0 the centrifugal term makes $u \approx r^{l+1}$. So put

$$u(r) = r^{l+1}e^{-\eta r}v(r).$$

By the product rule,

$$\frac{u''}{r^{l+1}e^{-\eta r}} = v'' + 2\Big(\frac{l+1}{r} - \eta\Big)v' + \Big(\frac{l(l+1)}{r^2} - \frac{2\eta(l+1)}{r} + \eta^2\Big)v.$$

Dividing $H_l u = Eu$ by $-\hbar^2/2m$, with $2mC/\hbar^2 = 2/a_0$, gives $u'' = (l(l+1)/r^2 - 2/a_0 r + \eta^2)u$. So $H_l u = Eu$ exactly when

$$rv'' + 2(l+1-\eta r)v' + 2\Big(\frac{1}{a_0} - \eta(l+1)\Big)v = 0. \tag{15.1}$$

For a series $v = \sum_k c_k r^k$, the coefficient of r^k gives

$$(k+1)(k+2l+2)\,c_{k+1} = 2\Big(\eta(k+l+1) - \frac{1}{a_0}\Big)c_k. \tag{15.2}$$

The series stops after the degree n_r exactly when $\eta(n_r + l + 1) = 1/a_0$. Then $\eta = 1/na_0$ with the *principal quantum number* $n = n_r + l + 1$, and $E = -\hbar^2/2mn^2a_0^2 = -\mathrm{Ry}/n^2$.

Theorem 15.1 (The levels of hydrogen). *Let $n \geq 1$ and $0 \leq l \leq n-1$ be integers, and let $v(r) = \sum_{k=0}^{n-l-1} c_k r^k$ with*

$$c_0 = 1, \qquad c_{k+1} = \frac{2(k+l+1-n)}{(k+1)(k+2l+2)\,na_0}\,c_k.$$

For every $p \in \mathcal{V}_l$, the function $\psi(x) = p(x)\,v(|x|)\,e^{-|x|/na_0}$ lies in $D(H)$, and

$$H\psi = -\frac{\mathrm{Ry}}{n^2}\,\psi.$$

The eigenspace of H for $-\mathrm{Ry}/n^2$ has dimension at least n^2.

Proof.

1. *The radial equation.* Let $\eta = 1/na_0$, so that $\eta(k+l+1) - 1/a_0 = (k+l+1-n)/na_0$, and put $c_k = 0$ for $k \geq n-l$. Then the c_k satisfy (15.2) for every $k \geq 0$, since for $k = n-l-1$ its right side vanishes. So the polynomial v satisfies (15.1) in every power of r, and $u = r^{l+1}e^{-\eta r}v$ has $H_l u = -(\mathrm{Ry}/n^2)u$.
2. *An eigenvector.* Since $\psi = up/r^{l+1}$, Proposition 13.8 gives $-(\hbar^2/2m)\Delta\psi = -(V + \mathrm{Ry}/n^2)\psi$ on $x \neq 0$. The function ψ, a polynomial in x and $|x|$ times $e^{-|x|/na_0}$, is bounded, continuous and square integrable. On $x \neq 0$ it is smooth, and its first partial derivatives are bounded, since $\partial_j |x| = x_j/|x|$ is bounded. Finally $V\psi$ is square integrable, since ψ is bounded and square integrable and $|x|^{-2}$ is integrable near 0, and so is $\Delta\psi$. By Lemma 14.3, $\psi \in D(H_0) = D(H)$ and $H_0\psi = -(\hbar^2/2m)\Delta\psi$. So $H\psi = -(\mathrm{Ry}/n^2)\psi$.
3. *The dimension.* Each $\mathcal{V}_l$ has dimension $2l+1$ (§ 13.4). Let $\varphi_l = p_l v_l e^{-|x|/na_0}$, with $p_l \in \mathcal{V}_l$ and v_l the polynomial for l, and suppose that $\sum_{l=0}^{n-1} \varphi_l = 0$. On $x \neq 0$ the L_j commute with functions of $|x|$ (Proposition 13.4), and $L^2 p_l = \hbar^2 l(l+1)p_l$, so $L^2\varphi_l = \hbar^2 l(l+1)\varphi_l$. Applying the product of the $L^2 - \hbar^2 l'(l'+1)$ over $l' \neq l$ to the sum leaves φ_l times a number that is not 0, since the $l(l+1)$ differ. So $\varphi_l = 0$. Since $v_l(0) = 1$, $v_l \neq 0$ near 0, so p_l vanishes near 0, and the polynomial p_l is 0. Hence the dimension is at least $\sum_{l=0}^{n-1}(2l+1) = n^2$. ∎

For $n = 1$, $v = 1$ and $\psi = \psi_1$ of Theorem 14.4. The polynomials v are Laguerre polynomials (Teschl, (10.65)).

Completeness and multiplicity

We next show that these eigenvectors span every eigenspace. Let ψ be an eigenvector of H. By Lemma 14.1, ψ is continuous. For $p \in \mathcal{V}_l$, its *radial coefficient* along p is

$$u_p(r) = r \int_{S^2} \overline{p(\hat{x})}\, \psi(r\hat{x})\, \mathrm{d}\Omega(\hat{x}), \qquad r > 0,$$

where $\hat{x}$ runs over the unit sphere S^2, with $\mathrm{d}\Omega = \sin\theta\, \mathrm{d}\theta\, \mathrm{d}\varphi$ (§ 13.4). The radial coefficient u_p depends antilinearly on p. A lemma from calculus will turn the eigenvalue equation into the radial equation for u_p, and a fact about polynomials will show that ψ is determined by its radial coefficients.

Lemma 15.2 (Weak solutions). *Let u and f be continuous on an open interval J, and suppose that $\int_J uq''\, \mathrm{d}r = \int_J fq\, \mathrm{d}r$ for every smooth q that vanishes outside a closed bounded interval inside J. Then u is twice continuously differentiable, and $u'' = f$.*

Proof. Fix $r_0 \in J$ and let $\Phi(r) = \int_{r_0}^{r} \int_{r_0}^{t} f(s)\, \mathrm{d}s\, \mathrm{d}t$, so that $\Phi'' = f$ and, integrating by parts twice, $\int \Phi q'' = \int fq$. So $w = u - \Phi$ has $\int wq'' = 0$ for every such q. Choose two such functions q_0 and q_1 with $\int q_0 = 1$, $\int rq_0 = 0$, $\int q_1 = 0$ and $\int rq_1 = 1$ (combinations of a bump and its translate). For every such q, the function $\tilde{q} = q - (\int q)q_0 - (\int rq)q_1$ has $\int \tilde{q} = \int r\tilde{q} = 0$. So $y(r) = \int_a^r \int_a^t \tilde{q}(s)\, \mathrm{d}s\, \mathrm{d}t$, with a below the supports, vanishes outside a closed bounded interval and has $y'' = \tilde{q}$, and $\int w\tilde{q} = \int wy'' = 0$. Hence $\int wq = A\int q + B\int rq$, that is, $\int (w - A - Br)q = 0$ for every q, with $A = \int wq_0$ and $B = \int wq_1$. A continuous function with this property vanishes, since otherwise its real or imaginary part has one sign on a small interval, where a bump q gives a nonzero integral. So $u = \Phi + A + Br$, which is twice continuously differentiable, with $u'' = \Phi'' = f$. ∎

Every polynomial is a sum of terms $|x|^{2j}\, p(x)$ with $p \in \mathcal{V}_l$. Indeed, let $N \geq 2$, and recall L13's Gaussian $g(x) = e^{-|x|^2/2s^2}$, for a fixed length $s > 0$. Multiplication by $|x|^2$ commutes with the L_j (Proposition 13.4) and maps $\mathcal{P}_{N-2}$ one-to-one into $\mathcal{P}_N$, so L^2 has on $|x|^2\, \mathcal{P}_{N-2}g$ the eigenvalues $\hbar^2 l(l+1)$, $l = N-2, N-4, \ldots$ (§ 13.4), and on $\mathcal{V}_N g$ the eigenvalue $\hbar^2 N(N+1)$. The two spaces meet only in 0, and their dimensions add up to that of $\mathcal{P}_N$, so $\mathcal{P}_N = \mathcal{V}_N \oplus |x|^2\, \mathcal{P}_{N-2}$. Since $\mathcal{P}_0 = \mathcal{V}_0$ and $\mathcal{P}_1 = \mathcal{V}_1$, induction on N gives the claim.

Theorem 15.3 (The spectrum of hydrogen). *The spectrum of H is*

$$\sigma(H) = \left\{ -\frac{\mathrm{Ry}}{n^2} : n = 1, 2, \ldots \right\} \cup [0, \infty), \qquad \sigma_{\mathrm{ess}}(H) = [0, \infty).$$

Every eigenvalue of H is one of the $-\mathrm{Ry}/n^2$, and its eigenspace has dimension n^2. It is spanned by the functions of Theorem 15.1.

Proof.

1. *The radial coefficients.* Let $H\psi = E\psi$ with $\psi \neq 0$. By Proposition 14.8, $E = -\hbar^2\eta^2/2m$ with $\eta > 0$. Let $p \in \mathcal{V}_l$. Since ψ is continuous and bounded, u_p is continuous, and $|u_p(r)| \leq r \sup|\psi| \int_{S^2} |p|\, \mathrm{d}\Omega$. By the Cauchy–Schwarz inequality on the sphere, $\int_0^\infty |u_p|^2\, \mathrm{d}r \leq \int_{S^2} |p|^2\, \mathrm{d}\Omega\, \|\psi\|^2$.
2. *The radial equation.* Let q be smooth and vanish outside a closed bounded interval inside $(0, \infty)$, and put $\zeta(x) = q(|x|)p(x)/|x|^{l+1}$. By Proposition 13.8, $\zeta \in \mathcal{S}(\mathbb{R}^3)$ and $H\zeta = (H_l q)(|x|)p(x)/|x|^{l+1}$. Integrating in spherical coordinates, $\langle \zeta, \psi \rangle = \int_0^\infty \bar{q} u_p\, \mathrm{d}r$,

and in the same way $\langle H\zeta, \psi\rangle = \int_0^\infty \overline{H_l q}\, u_p \,\mathrm{d}r$. Since $\langle H\zeta, \psi\rangle = \langle \zeta, H\psi\rangle = E\langle \zeta, \psi\rangle$, dividing by $-\hbar^2/2m$ gives, for real q, and so by linearity for every q,

$$\int_0^\infty q'' u_p \,\mathrm{d}r = \int_0^\infty q\left(\frac{l(l+1)}{r^2} - \frac{2}{a_0 r} + \eta^2\right) u_p \,\mathrm{d}r.$$

By Lemma 15.2, u_p is twice continuously differentiable and solves the radial equation $H_l u_p = E u_p$.

3. *The regular solution.* Let $c_0 = 1$ and define the c_k by (15.2). Either the series stops, or every c_k is nonzero and $c_{k+1}/c_k \to 0$. In both cases $v = \sum_k c_k r^k$ converges for every r, can be differentiated term by term, and $u_1 = r^{l+1} e^{-\eta r} v$ solves the radial equation. The Wronskian $W = u_1 u_p' - u_1' u_p$ is constant, since $W' = u_1 u_p'' - u_1'' u_p = 0$. Since $u_1(r)/r^{l+1} \to v(0) = 1$ as $r \to 0$, we can choose $\epsilon \in (0,1)$ with $r^{l+1}/2 \le u_1(r) \le 2r^{l+1}$ on $(0,\epsilon]$. Suppose $W \neq 0$. On $(0,\epsilon]$, $(u_p/u_1)' = W/u_1^2$, so

$$u_p(r) = u_1(r)\left(\frac{u_p(\epsilon)}{u_1(\epsilon)} - W\int_r^\epsilon \frac{\mathrm{d}s}{u_1(s)^2}\right), \qquad \int_r^\epsilon \frac{\mathrm{d}s}{u_1(s)^2} \ge \frac{r^{-2l-1} - \epsilon^{-2l-1}}{4(2l+1)},$$

and $\left|u_p(r)\right| \ge b\, r^{-l}$ for small r, with some $b > 0$. This contradicts the bound $\left|u_p(r)\right| \le \text{const}\cdot r$ of the first step. So $W = 0$. Let $\beta = u_p(\epsilon)/u_1(\epsilon)$. The solution $w = u_p - \beta u_1$ has $w(\epsilon) = 0$ by the choice of β, and its Wronskian with u_1 is $u_1 w' - u_1' w = W = 0$. At ϵ this gives $u_1(\epsilon) w'(\epsilon) = 0$, and $u_1(\epsilon) > 0$, so $w'(\epsilon) = 0$. Hence $w = 0$ by the proof of Lemma 10.6 on each closed bounded interval inside $(0,\infty)$. So $u_p = \beta u_1$.

4. *Square integrability.* If the series does not stop, every c_k is nonzero, and $(k+1)c_{k+1}/c_k \to 2\eta$. So for $k \ge K$ the c_k have one sign and $|c_{k+1}| \ge (3\eta/2)\,|c_k|\,/(k+1)$. Then

$$|v(r)| \ge |c_K|\; K!\left(\frac{3\eta}{2}\right)^{-K}\left(e^{3\eta r/2} - \sum_{k<K}\frac{(3\eta r/2)^k}{k!}\right) - \left|\sum_{k<K} c_k r^k\right|,$$

and u_1 grows at least like $e^{\eta r/2}$, so it is not square integrable. Since u_p is, $u_p = 0$ unless the series stops, that is, unless $\eta = 1/na_0$ with an integer $n \ge l+1$.

5. *All coefficients zero.* Suppose $u_p = 0$ for every l and every $p \in \mathcal{V}_l$. For $p \in \mathcal{V}_l$ and $j \ge 0$, with L13's Gaussian g, integrating in spherical coordinates, $\left\langle |x|^{2j} pg, \psi\right\rangle = \int_0^\infty r^{2j+l+1} e^{-r^2/2s^2} u_p(r)\,\mathrm{d}r = 0$. By the decomposition of polynomials above, ψ is orthogonal to every $Q_N g$, so to every $h_{n_1 n_2 n_3}$, and $\psi = 0$ by Theorem 13.5.

6. *The eigenvalues.* If E is not one of the $-\mathrm{Ry}/n^2$, then by the fourth step every $u_p = 0$, and $\psi = 0$ by the fifth. So every eigenvalue is one of the $-\mathrm{Ry}/n^2$. For $E = -\mathrm{Ry}/n^2$, the fourth step gives $u_p = 0$ for $l \ge n$, and for $l \le n-1$ the third gives $u_p = \beta(p) u_{nl}$ with $u_{nl} = r^{l+1} e^{-r/na_0} v$ of Theorem 15.1. Choose a basis $(p_{l,i})$ of each $\mathcal{V}_l$, $l \le n-1$. By the fifth step, the linear map $\psi \mapsto (\beta(p_{l,i}))_{l,i}$ from the eigenspace to $\mathbb{C}^{n^2}$ is one-to-one. So the eigenspace has dimension at most n^2, and by Theorem 15.1 exactly n^2, spanned by the functions found there.

7. *The spectrum.* By Theorem 14.7, every point of $\sigma(H)$ below 0 is an eigenvalue, and every $-\mathrm{Ry}/n^2$ is one. With Proposition 14.5 and Theorem 14.7, this is the claim. ∎

Teschl, Thm 10.9 (with his Thms 10.2, 10.3 and 10.8), obtains the negative eigenvalues and their multiplicities by separating the angular variables first. Hall, Thm 18.4 and Cor. 18.5, sketches the same route and cites the discreteness of the negative spectrum. Lecture 13 asked which self-adjoint operator on $L^2(0,\infty)$ the Hamiltonian becomes on the functions $(u(r)/r)Y_{lm}$, and on which domain. The proof above does not need the answer, which we state.

Theorem 15.4 (The sectors · *cited*). *Let H be H_0, or $H_0 - C/|x|$ with $C > 0$, on $D(H_0)$. For each l and μ, the projection onto the vectors $(u(r)/r)Y_{l\mu}$ maps $D(H)$ into itself and commutes with H, and in terms of u, H acts on these vectors as a self-adjoint operator A_l on $L^2(0,\infty)$: $A_l u = H_l u$ on the functions $u \in L^2(0,\infty)$ with u and u' absolutely continuous on every closed bounded interval inside $(0,\infty)$ and $H_l u \in L^2(0,\infty)$, and, for $l = 0$, with* $\lim_{r\to 0}(u(r) - ru'(r)) = 0$.

This is Teschl, (10.45)–(10.50) and Thm 10.8, in units with $\hbar^2/2m = 1$. For a function u continuously differentiable on $[0,\infty)$, the condition for $l = 0$ says $u(0) = 0$. It excludes $u = e^{-\eta r}$, the radial function of the guard $e^{-\eta|x|}/|x|$ in § 14.2. In the proof above, the bound $|u_p(r)| \le \text{const} \cdot r$ did that work: it excluded the second solution, which near 0 is at least a multiple of r^{-l} in absolute value, a constant for $l = 0$.

Self-check 15.1

1. Use the recursion to find v for $n = 3$ and $l = 0$.

15.2 The degeneracy n^2

The level $-\text{Ry}/n^2$ holds the states of angular momentum $l = 0, 1, \dots, n-1$, with $2l+1$ independent states for each l and $\sum_{l=0}^{n-1}(2l+1) = n^2$ in all. Spectroscopy names $l = 0, 1, 2, 3$ by the letters s, p, d, f, so the level $n = 2$ holds one $2s$ state and three $2p$ states, and the level $n = 3$ holds $3s$, $3p$ and $3d$, with $1 + 3 + 5 = 9$ states.

Rotations explain part of this. Each $U(R)$ commutes with multiplication by V, which depends only on $|x|$. By the self-check of § 13.1 on $\mathcal{S}(\mathbb{R}^3)$ and the first and third claims of Lemma 14.1, each $U(R)$ maps $D(H_0)$ into itself and commutes with H_0. So the eigenspaces of H are kept by the rotations. Each is a sum of whole multiplets (Hall, p. 397), as *Quantum Mechanics I*, Cor. 7.5, leads one to expect of a Hamiltonian that commutes with rotations. That forces the $2l+1$ states of one l, and nothing more. That the angular momenta $l = 0, 1, \dots, n-1$ share one level is not forced by rotations. As in the oscillator of § 13.6, it comes from a larger symmetry. Optional: Hall, Thms 18.14 and 18.16, explains this symmetry using the Runge–Lenz vector. Nothing here depends on that construction.

Self-check 15.2

1. How many independent states of H have an energy below $-\text{Ry}/16$? Which angular momenta l occur at the level $n = 3$?

15.3 The Balmer lines

An atom in the level n can fall to a lower level n' and emit a photon, which carries the difference of the energies. This is a rule of the coupling to light, not derived here, as in the physics box of Lecture 13. A photon of energy E has the wavelength $\lambda = 2\pi\hbar c/E$, with the speed of light c.

Example 15.5 (The Balmer lines).

1. *The energies.* A fall from $n \ge 3$ to $n' = 2$ emits $\text{Ry}(1/4 - 1/n^2)$. For $n = 3$ this is $5\text{Ry}/36 \approx 1.89\,\text{eV}$.
2. *The wavelengths.* With the *Rydberg constant* $R_\infty = \text{Ry}/2\pi\hbar c \approx 1.0974 \times 10^7\,\text{m}^{-1}$,

$$\frac{1}{\lambda} = R_\infty\left(\frac{1}{4} - \frac{1}{n^2}\right).$$

For $n = 3, 4, 5, 6$ this gives 656.1, 486.0, 433.9 and 410.1 nm, four lines in the visible: red, blue-green and two violet.

3. *The measurement.* The lines of hydrogen lie at 656.3, 486.1, 434.0 and 410.2 nm (NIST, in air). Each prediction is 0.03% short. The proton moves too. With the reduced mass $mm_p/(m+m_p)$ in place of m, the wavelengths grow by 0.05%. The air shortens them by 0.03%.

Self-check 15.3

1. A fall from $n = 2$ to $n' = 1$ emits the Lyman-α line. Find its wavelength.

The physics

Balmer found the rule $1/\lambda \propto \frac{1}{4} - 1/n^2$ in 1885 from four lines measured by Ångström. Here it follows from the operator H with its domain, and R_∞ from $\hbar$, m, e, ε_0 and c. The intrinsic spins of the electron and the proton, left out here, split each line further. The electron's spin couples to its orbital angular momentum, as in the doublet of sodium (*Quantum Mechanics I*, §13.6). The proton's spin splits the ground state into the two levels of the 21 cm line of *Quantum Mechanics I*, Lecture 10. That splitting is proportional to the density of the electron at the proton, $|\psi_1(0)|^2 / \|\psi_1\|^2 = 1/\pi a_0^3$.

15.4 What a continuous quantum system is

This course asked the four questions of *Quantum Mechanics I* for a particle, and the continuum changed each answer.

States. A state is the ray of a unit vector in a space of infinite dimension, L^2 of an interval, of the line or of space. Its vectors are classes of functions, and a position has no eigenvector in it.

Observables. An observable is a self-adjoint operator, with its domain. A symmetric operator is not enough. The momentum P_D in a box is symmetric and not self-adjoint, and its self-adjoint versions P_θ give different readings (Lecture 3). Nor is a formula enough. Hydrogen's rule $-(\hbar^2/2m)\Delta - C/|x|$ is self-adjoint on $D(H_0)$ with the least energy −Ry, and its other self-adjoint extensions from the smooth functions that vanish near the origin each have a level below −Ry (§ 14.2). The spectrum of a self-adjoint operator is the set of numbers with approximate eigenvectors (Theorem 6.5). It is discrete in the box and the oscillator, continuous for the free particle, and has both parts in the finite well and in hydrogen.

Measurement. A reading asks whether an observable lies in an interval, and the spectral measure gives its probability (Postulate 5.5).

Evolution. Evolution is a strongly continuous unitary group, and its generator is self-adjoint (Stone, Theorem 7.8). For a potential on the line, the states of $\mathcal{H}_{\text{pp}}$ stay near the well, and every state $\Omega_{\text{out}}\chi$ leaves every bounded interval (Theorem 12.6).

Foundations takes up what this course left out: states that are density operators on an infinite-dimensional space, readings more general than spectral measures, and the question of what an observation is.

Hydrogen

States: the levels $-\text{Ry}/n^2$ have the eigenvectors $p(x)v(|x|)e^{-|x|/na_0}$, $p \in \mathcal{V}_l$, $l \le n-1$ (Theorem 15.1), and these span each eigenspace (Theorem 15.3). *Observables:* the

energy $H = H_0 - C/|x|$, self-adjoint on $D(H_0)$ (Theorem 14.2), has the spectrum $\{-\mathrm{Ry}/n^2\} \cup [0, \infty)$, and each level holds exactly n^2 independent states (Theorem 15.3). *Measurement:* a reading of the energy in an interval that misses $\{-\mathrm{Ry}/n^2\} \cup [0, \infty)$ answers yes with probability 0 (Theorem 15.3, and Theorem 6.6, cited).

Summary.

- *The levels of hydrogen*: $-\mathrm{Ry}/n^2$, each with exactly n^2 independent eigenvectors, and no other eigenvalue (Theorems 15.1 and 15.3).
- *The degeneracy*: the level $-\mathrm{Ry}/n^2$ holds the angular momenta $l = 0, 1, \ldots, n-1$. Rotations force the $2l+1$ states of each l, and not that the l share one level (§ 15.2).
- *The Balmer lines*: $1/\lambda = R_\infty(1/4 - 1/n^2)$, within 0.03% of the measured lines (Example 15.5).

The end of the course. Foundations recalls these answers and asks what they leave open.

Self-check answers.

15.1 1. Here $c_1 = -2/3a_0$ and $c_2 = c_1(-2)/(2 \cdot 3 \cdot 3a_0) = 2/27a_0^2$, so $v = 1 - 2r/3a_0 + 2r^2/27a_0^2$.

15.2 1. The levels $n = 1, 2, 3$, with $1 + 4 + 9 = 14$ states. At $n = 3$, $l = 0$, 1 and 2.

15.3 1. $1/\lambda = R_\infty(1 - \frac{1}{4})$, so $\lambda \approx 121.5\,\mathrm{nm}$, in the ultraviolet.

References

Teschl, *Mathematical Methods in Quantum Mechanics*, Thms 10.2 and 10.3 (pp. 222–223), (10.45)–(10.50) and Thm 10.8 (pp. 229–230), Thm 10.9 (pp. 231–233). Hall, *Quantum Theory for Mathematicians*, Thms 18.3 and 18.4 (pp. 397–400), Cor. 18.5 (p. 401), Thms 18.14 and 18.16 (pp. 406–407). NIST Atomic Spectra Database, lines of H I.